

Hospital Resource Stability Lab

Brief Project Report — Pandemic Healthcare Stability Analysis

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Phase 1: Banker's + Monte Carlo | Phase 2: C(t), LOS, overflow
Phase 3: Public-data calibration (A/B/C models)

What this is

- An academic simulation of hospital resource allocation under pandemic-like pressure.
- It answers: (1) Is the current allocation state SAFE? (Banker's Algorithm) and
- (2) As patient load grows, what is the probability of collapse? (Monte Carlo).
- This is not a live hospital system or clinical decision tool. Results are model-based.

Core model (Phase 1)

- Five abstract resources: ICU beds, oxygen, ventilators, nurses, blood units.
- Each patient has Allocation, Max demand, Need = Max - Allocation, and Available capacity.
- Banker's Algorithm (Dijkstra, 1965) returns SAFE if a completion ordering exists, else UNSAFE.
- Monte Carlo sweeps $n = 1..N$ patients with Wilson 95% CIs on $P(\text{collapse})$.
- Collapse = admission failure OR unsafe Banker's state.
- Isolated sensitivity: elasticity $\epsilon_{k,C} = (n_{50}(C') - n_{50}(C)) / \Delta C_k$ per resource.

Time-dependent hospital (Phase 2)

- Discrete time (default 1 hour): $C(t)$ for nurse staffing (shift/weekend multipliers).
- Length of stay, fatigue (high utilization extends LOS), treatment bundles (all-or-nothing).
- Overflow state (ED_OVERFLOW) when bundles cannot be satisfied — not automatic collapse.
- Acuity degradation in overflow; severity-based triage for admission order.
- Banker's SAFE/UNSAFE remains a separate per-timestep snapshot. Preemption not implemented.

Empirical calibration (Phase 3)

- Public data pipeline: Fetch -> Validate -> Clean -> Cache -> Calibrate (with fallback).
- Sources: OWID COVID-19, UK Coronavirus Dashboard, Rees et al. 2020 LoS review.
- Every quantity labeled: REAL OBSERVATION, MODEL ASSUMPTION, DERIVED PARAMETER, or SIMULATED.
- Arrivals: Poisson vs negative binomial (overdispersion tested). LOS: Gamma vs Weibull vs Lognormal.
- Model A = independent synthetic; B = severity-conditioned joint ICU/vent; C = calibrated.
- Copulas not used (no patient-level resource microdata).

Dashboard and verification

- Web UI: Banker's trace, Monte Carlo curve, sensitivity, time model, Phase 3 A/B/C comparison.
- CLI: `python pandemic_bankers/main.py`. Tests: `python tests/run_tests.py` (68 tests).
- Mathematics reference page documents formulas and assumptions.

Limitations (for reviewers)

- Five abstract resources; max demand assumed known (Banker's requirement).
- National/epidemiological data are not this hospital's live census.
- Under surge capacity, $P(\text{unsafe})$ is often near zero; admission failure dominates.
- Phase 2 parameters (fatigue, shifts) are sensitivity knobs unless backed by evidence.
- Not clinically validated — illustrative for allocation theory and stochastic risk.

One-sentence pitch

- A reproducible academic lab combining Banker's deadlock avoidance, Monte Carlo collapse
- risk, discrete-time hospital dynamics, and optional public-data calibration — with clear
- separation between observations, assumptions, and simulation.

Questions for feedback

- Are the five resources and demand profiles plausible for your setting?
- Is overflow vs Banker's UNSAFE the right split for system stress vs deadlock?
- Should Phase 3 prioritize local hospital CSV upload over national APIs?
- Which capacity scenario (baseline vs surge) best matches your narrative?
- Is the A/B/C experiment the right comparison for a paper?